Optimising water delivery under constraints

Overview and context

Challenge description

The access to clean water has been recognised as a critical challenge for humanity, with many communities receiving far below the WHO-recommended minimum of 100L per person and per day. More than 2 billion people (1 in 4) still live without safely managed drinking water [1].

In several regions, limited infrastructure, over-extraction of aquifers, and energy constraints make water supply planning complex. Geopolitical factors further exacerbate scarcity. Under that lens, efficient and equitable allocation of scarce water resources is a fundamental and persistent challenge, critical to support public health, agriculture, and sustainable development. Notably, ensuring equitable water access directly supports the United Nations' Sustainable Development Goals 6, 7, and 13.

While classical algorithms are a successful and well-established tool, quantum-computational approaches might provide a more performant framework for addressing such combinatorial resource allocation problems, especially at scale and under multifaceted constraints.

This challenge proposes an exploration of how quantum algorithms may constitute an appealing alternative to tackle combinatorial constrained optimisation problems. Variational quantum algorithms, such as QAOA[2] or VQE[3], can be used to identify optimal or near-optimal allocation strategies.

Link with the UN's sustainable development goals



SDG 6 — Clean Water and Sanitation
Directly about optimising water supply to communities. Addresses access to safe water, efficient use of scarce water resources, and distribution under constraints.

SDG 7 — Affordable and Clean Energy
Energy limits are part of the problem. Optimising energy use in pumping and desalination promotes sustainable energy management.





SDG 13 — Climate Action
Efficient water–energy management helps adapt to arid conditions and climate variability. Reduces energy waste and helps mitigate resource-related stress in a climate-sensitive region.

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Technical elements

Problem mapping

The problem of water allocation can be mathematically described as a combinatorial optimisation problem. We model the system using a small number of water sources, including wells, desalination plants, or purchased water links. Each source is represented by a binary decision qubit indicating whether it is operational. Communities or demand regions can also be represented as qubits to track fulfillment of minimum water requirements.

The objective is to maximise the total water delivered while respecting energy and sustainability constraints, which include limits on simultaneous plant operation, sustainable extraction thresholds, and minimum supply targets. These constraints are encoded as penalty terms in a QUBO or Hamiltonian formulation.

Mathematical modeling

The problem can be represented as a graph, where each water source and each region is represented by a node. Whether a source is able to convey water to a given region, and the daily volume it can provide, is encoded in a weighted edge between the corresponding nodes (Figure 1).

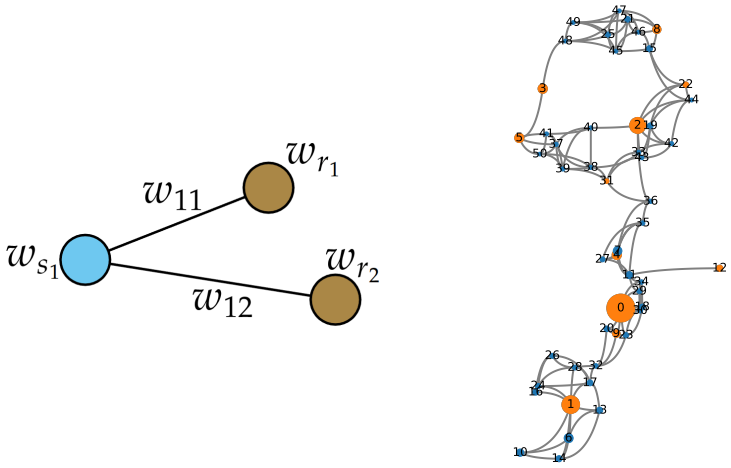


Figure 1: Left: A simple graph representing a source and two demand regions. The source is able to produce a volume w_{s_1} of water per day, and convey a volume w_{11} (resp w_{12}) to the region 1 (resp 2),. The weights w_{r_1}, w_{r_2} correspond to the amount of water needed by each region. Right: a realistic graph with 51 nodes. Orange nodes correspond to administrative centres of the corresponding regions. Sources are not represented.

The state of each water source can be encoded in the state of a qubit ($|0\rangle$: inactive, $|1\rangle$: active).

The goal is to maximise the water delivery to all sites:

$$\max_{\vec{x}} \sum_{\substack{s\in \text{ Sources} \\ r\in \text{ Regions}}} w_{sr} x_s x_r$$

with the following possible constraints:

- Each source can deliver a maximum volume of water w_s:

$$\sum_{r\in \text{ Regions}} w_{sr} x_r \leq w_s x_s.$$

- Ideally, each consumption region should receive a minimum quantity of water w_r:

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$$\sum_{r\in \text{ Sources}} w_{sr} x_s \geq w_r x_r.$$

- Only a limited number of sources $S_{\max}$ can be active at the same time, e.g. due to energetic constraints:

$$\sum_{s\in \text{ Sources}} x_s \leq S_{\max}$$

With the goal of maximising the quantity of water supplied to the whole population under the specified constraints, the objective function to be *minimised* can be written

$$\begin{aligned} H = &- \sum_{\substack{s\in \text{ Sources} \\ r\in \text{ Regions}}} w_{sr} x_s x_r \\ &+ \sum_{s\in \text{ Sources}} \lambda_s \left(-w_s x_s + \sum_{r\in \text{ Regions}} w_{sr} x_r \right) \\ &+ \sum_{r\in \text{ Regions}} \lambda_r \left(+w_r x_r - \sum_{s\in \text{ Sources}} w_{sr} x_s \right) \\ &+ \mu \left(S_{\max} - \sum_{s\in \text{ Sources}} x_s \right)^2. \end{aligned}$$

The parameters $\{\lambda_s, \lambda_r, \mu\}$, called Lagrange multipliers, aim at enforcing the satisfaction of the constraints by penalising strong deviations. In theory, appropriate values can be determined by choosing larger values than the maximum possible gain from deviations. In practice, this is often done empirically due to the interplay of the different constraints. Finally, whether the constraints are enforced with linear or quadratic penalties is based on physical interpretation; note that other forms are possible (e.g. $\max(0, \text{value-constraint})$) and worth exploring.

Complexity reduction and reduction to standard problems

Single-multipliers assumptions

First, nobody wants to handle dozens of Lagrange multipliers: we can with more or less safety — depending on the graph— choose a single multiplier per constraint type, e.g. $\lambda_S = \max_{s\in \text{ Sources}} \lambda_s$ and $\lambda_R = \max_{r\in \text{ Regions}} \lambda_r$. This reduces the number of multipliers from $S + R + 1$ to only 3 (which is still larger than 1).

NP-hardness and reduction to standard optimisation problems

Max-Cut

If we assume (1) no constraint on the maximum number of active sources, (2) large source capacity, and (3) low demand, then all constraints can be ignored, leaving only

$$H = - \sum_{\substack{s\in \text{ Sources} \\ r\in \text{ Regions}}} w_{sr} x_s x_r,$$

which is exactly a Max-Cut problem (one can do for instance do the change of variables $x_r \to 1 - x_r$ to get to the more common form). Therefore, the full optimisation problem generalises Max-Cut. Since the latter is NP-hard, our problem is NP-hard itself.

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Pros:

- No constraints. This makes the problem a bit simpler to implement since there are no Lagrange multipliers to adjust.

Cons:

- Not realistic in the given context: neither (1), (2), nor (3) correspond to the actual situation.
- Realistically, all regions will be active. This reduce the number of variables to only the ones corresponding to sources.

Knapsack

We can (1) fix the demand regions to be always active ($x_r = 1$), assume (2) large source capacity, and (3) low demand. This keeps only the constraint that the total number of sources is limited, turning the problem into maximising

$$\sum_{s \in \text{ Sources}} x_s \left(\sum_{r \in \text{ Regions}} w_{sr} \right)$$

with the constraint

$$\sum_{s \in \text{ Sources}} x_s \leq S_{\max}.$$

This time, we end up with a knapsack formulation,which is again NP-hard. This reduction too proves that the full problem is NP-hard.

Pros:

- Slightly more realistic than Max-Cut.

Cons:

- Still not realistic in the given context: neither (2) nor (3) correspond to the actual situation.

Going further

The present exposition assumes a binary model: sources/regions are either active or inactive. A more realistic modelling should consider partial flows of water. How would you approach this problem?

Dataset

Two datasets are provided:

- One represents the 11 administrative regions of a territory. The GPS coordinates are those of the main city of each region, and the populations are the total estimated population within said region.
- One represents the 51 most populated cities of the territory.

For both, edges between the cities are generated within the code. These links do not reflect actual infrastructure; rather, they are generated in a way that makes the problem more pedagogical.

References

[1] United Nation's *Water Facts*, https://www.unwater.org/water-facts (2025)

[2] K. Blekos et al., *A Review on Quantum Approximate Optimization Algorithm and its Variants* (2023).

[3] J. Tilly et al., *The Variational Quantum Eigensolver: a review of methods and best practices* (2021)

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